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Unity3D Programmer | Game Architect & Technical 3D Artist

Unity/C# developer with sprint-cycle delivery and sole technical lead on a 3D Android mobile game with five distinct gameplay mechanics. Built a custom OpenGL rendering engine from scratch MVP matrices, Phong lighting, no external math library.

CORE COMPETENCIES & CERTIFICATIONS

Programming: C#, C++, Python, object-oriented programming, data structures, algorithm design, algebra, geometry, trigonometry, vectors, and matrices.

Tools and Platforms: Unity, Unreal Engine, OpenGL, Adobe Photoshop, Visual Studio, Git, GitHub, Blender.

AI and automation: Unity ML-Agents, finite state machines, decision trees, ONNX integration, Python scripting, and third-party integration (e.g., Multiplayer(Photon)/Database(mySQL)).

Delivery & Execution: Agile teamwork, debugging, testing, performance optimization, documentation, designing scalable architectures, and project tracking with Jira.

Unity Modules: Animator, Physics 2D/3D, Cinemachine, Unity UI/TextMeshPro, Audio Mixer, Addressables, Unity Profiler, NavMesh.

PROFESSIONAL EXPERIENCE

Jul-Sept 2025

[Game Developer Intern, WRP, Islamabad](#)

- **Shipped Unity/C# features** across 5 sprint cycles without a missed deadline, writing modular, component-based systems reviewed by peers and merged directly into the live product branch.
- **Optimized** the CyberSafe mobile build pipeline (IL2CPP, GPU instancing, static batching) using Unity Profiler to achieve a stable 30fps on mid-range Android devices while balancing visual fidelity.
- **Identified and resolved 25+ bugs** across five sprint cycles using systematic QA methodology (Unity Debugger/console logs), maintaining zero regressions on all integrated features.
- **Architected and delivered** a complete enemy patrol AI system in a single 2-week sprint, using a C# Finite State Machine for state transitions and integrated Unity's NavMesh Agent for dynamic pathfinding.

Selected Projects

2025-2026

[CyberSafe — 3D Mobile RPG, Unity 6, C#, Blender](#)

Led development of CyberSafe as the primary Unity/C# developer built five distinct gameplay minigames (Malware Maze, Phishing simulation, Password Strength, DDoS defence, and Ransomware), core scene architecture, mobile input handling, and the full Android build pipeline from project settings to APK export.

- **Configured** the complete Android build pipeline (IL2CPP, ARM64, static batching, GPU instancing) and profiled with Unity Profiler to achieve stable 30fps on mid-range Android devices (3GB+ RAM) balancing visual fidelity with hardware constraints.
- **Built a real-time password strength validator in C#** character-by-character scoring logic, integrated with Unity's mobile soft keyboard, with instant colour-coded UI feedback one of five distinct interactive mechanics across CyberSafe's episodic cybersecurity curriculum.
- **Found and fixed 10+ bugs** across the full project lifecycle using Unity Profiler, device logs, and reproduction testing covering scene transitions, mobile input edge cases, and build-time errors.
- **Integrated Unity Analytics** to track player progress events across all five chapters monitoring chapter completion rates and session behaviour.

2024

[Custom 3D Rendering Engine, C++, OpenGL](#)

- **Built a real-time 3D rendering engine in C++/OpenGL from the ground up:** implemented an OBJ asset loader, scene graph, orbit/pan camera controller, and a full rendering loop no engine, no rendering library, just the OpenGL API and linear algebra.
- **Wrote every line of 3D math by hand in C++:** model/view/projection matrix pipeline, Phong lighting model (ambient, diffuse, specular), and perspective-correct viewport built without GLM or any math library, demonstrating direct application of linear algebra to real-time rendering.

2024

[AI-Driven NPC Systems \(2024, Unity, C#\)](#)

- **Designed scalable enemy behaviour using a C# finite state machine** layered with decision tree logic enabling enemies to shift between rule-based patrol and adaptive decision-making based on player proximity and game state.

2024

[Football ML Simulation \(2024, Unity, C#, Python, ML-Agents, ONNX\)](#)

- **Trained agent behaviour using Unity ML-Agents with Python** scripting and YAML reward shaping, then exported via ONNX and integrated into Unity runtime for real-time inference with no Python dependency at runtime.

2023-2024

[Zombie Survivor — 3D Action-Survival Game, Unity, C#, Blender](#)

- **Architected the full gameplay loop in Unity/C#:** a collectible weapon unlock system (weapons hidden in-world and activated on pickup), health/damage model, score tracking, and game state manager all integrated with Unity Audio Mixer for sound feedback and TextMeshPro for in-game HUD.
- **Built the combat and AI perception systems together:** Physics.Raycast for hitscan weapon detection, trigger colliders for enemy awareness radii all event-driven inputs into a C# FSM governing patrol → chase → attack state transitions and back.
- **Built the full 3D art pipeline solo:** modelled environments and props in Blender, applied PBR textures (albedo, normal, roughness), rigged and animated the player character and zombie enemies, then exported via FBX into Unity with Animator Controller state transitions for movement and combat.

2023	Astro Rush (2023, Unity, C#) Built Astro Rush end-to-end in Unity/C#: a Rigidbody-driven vehicle controller tuned for realistic handling adjusted mass, drag, and angular drag to simulate weight and momentum alongside multi-level progression, obstacle generation, and planetary environment dressing.
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EDUCATION

2022-2026	Bachelor in Computer Games Development (BSCGD) E-9 Air University, Islamabad
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LANGUAGES

Urdu	Native Proficiency
English	Professional Working Fluency